

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Rackspace Richardson DFW3, TX -- station KDAL, America/Chicago
Committed pair OSM 597876726 -> 597876721
Digital Realty Dallas DFW35 / Digital Realty Dallas DFW17
Facade gap 176.5 m between the two halls
Weather record 43,803 real hourly records from KDAL
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.2257 C at 130 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2022-01-01 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point gate off

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 10 of 24 hours with 1 mode change. That is 9 more than the reactive incumbent, at 0 unsafe hours against its 1. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 4 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 10 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 1 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
the incumbent broke its own switch budget 1 time(s) to stay safe
4 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dry-bulb	18.155	18.0	18.330	0.788
01	mechanical	no	dry-bulb	18.403	18.0	18.330	0.950
02	mechanical	no	dry-bulb	18.351	18.0	17.884	0.992
03	mechanical	no	dry-bulb	18.389	18.0	17.781	0.873
04	mechanical	yes	switch budget	17.887	18.0	17.300	0.776
05	mechanical	yes	switch budget	17.521	18.0	16.774	0.780
06	mechanical	yes	switch budget	17.643	18.0	16.671	0.878
07	mechanical	yes	switch budget	17.621	18.0	16.834	0.743
08	mechanical	no	dry-bulb	18.744	18.0	16.834	1.360
09	mechanical	no	dry-bulb	20.143	18.0	17.300	1.684

10	mechanical	no	dry-bulb	21.801	18.0	19.441	1.675
11	mechanical	no	dry-bulb	22.295	18.0	20.560	1.567
12	mechanical	no	dry-bulb	22.843	18.0	21.670	1.460
13	mechanical	no	dry-bulb	24.001	18.0	22.780	1.273
14	FREE	yes	-	17.847	18.0	10.560	1.046
15	FREE	yes	-	15.659	18.0	6.110	0.955
16	FREE	yes	-	14.255	18.0	3.330	0.887
17	FREE	yes	-	6.998	18.0	2.220	0.879
18	FREE	yes	-	3.437	18.0	0.560	0.907
19	FREE	yes	-	1.170	18.0	-0.610	1.032
20	FREE	yes	-	0.104	18.0	-1.720	1.221
21	FREE	yes	-	-1.326	18.0	-2.780	1.099
22	FREE	yes	-	-2.467	18.0	-3.890	0.952
23	FREE	yes	-	-3.581	18.0	-5.000	0.819

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.155 C against a 18.0 C limit, so it fails by 0.155 C. It would take a limit 0.155 C higher, or a bound 0.155 C tighter, to change this hour.

01:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.403 C against a 18.0 C limit, so it fails by 0.403 C. It would take a limit 0.403 C higher, or a bound 0.403 C tighter, to change this hour.

02:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.351 C against a 18.0 C limit, so it fails by 0.351 C. It would take a limit 0.351 C higher, or a bound 0.351 C tighter, to change this hour.

03:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.389 C against a 18.0 C limit, so it fails by 0.389 C. It would take a limit 0.389 C higher, or a bound 0.389 C tighter, to change this hour.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.887 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.521 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.643 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.621 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.744 C against a 18.0 C limit, so it fails by 0.744 C. It would take a limit 0.744 C higher, or a bound 0.744 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.143 C against a 18.0 C limit, so it fails by 2.143 C. It would take a limit 2.143 C higher, or a bound 2.143 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.801 C against a 18.0 C limit, so it fails by 3.801 C. It would take a limit 3.801 C higher, or a bound 3.801 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.295 C against a 18.0 C limit, so it fails by 4.295 C. It would take a limit 4.295 C higher, or a bound 4.295 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.843 C against a 18.0 C limit, so it fails by 4.843 C. It would take a limit 4.843 C higher, or a bound 4.843 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.001 C against a 18.0 C limit, so it fails by 6.001 C. It would take a limit 6.001 C higher, or a bound 6.001 C tighter, to change this hour.

14:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.847 C, which is 0.153 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.046 C, and the margin is measured, not chosen: 0.979 C of group-conditional forecast error for this hour of day, plus 0.0670 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

15:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.659 C, which is 2.341 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.955 C, and the margin is measured, not chosen: 0.851 C of group-conditional forecast error for this hour of day, plus 0.1043 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.255 C, which is 3.745 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.887 C, and the margin is measured, not chosen: 0.797 C of group-conditional forecast error for this hour of day, plus 0.0904 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.998 C, which is 11.002 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.879 C, and the margin is measured, not chosen: 0.826 C of group-conditional forecast error for this hour of day, plus 0.0530 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 3.437 C, which is 14.563 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.907 C, and the margin is measured, not chosen: 0.805 C of group-conditional forecast error for this hour of day, plus 0.1016 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 1.170 C, which is 16.830 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.032 C, and the margin is measured, not chosen: 0.947 C of group-conditional forecast error for this hour of day, plus 0.0848 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.104 C, which is 17.896 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.221 C, and the margin is measured, not chosen: 1.086 C of group-conditional forecast error for this hour of day, plus 0.1342 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -1.326 C, which is 19.326 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.099 C, and the margin is measured, not chosen: 1.034 C of group-conditional forecast error for this hour of day, plus 0.0643 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -2.467 C, which is 20.467 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.952 C, and the margin is measured, not chosen: 0.894 C of group-conditional forecast error for this hour of day, plus 0.0583 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -3.581 C, which is 21.581 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.819 C, and the margin is measured, not chosen: 0.765 C of group-conditional forecast error for this hour of day, plus 0.0544 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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